

Mathematics 629
Spring 2026
Homework Assignment 6

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Due: 3/25/2026

1. Write out the complete details of the proof of the “differentiation under the integral” Theorem 1.129 in the notes.

Proof. i) Suppose there exists $g \in L^1(\mu)$ such that $|f(x, t)| \leq g(x)$ for all (x, t) and $\lim_{t \rightarrow t_o} f(x, t) = f(x, t_o)$. Let $f_n(x) = f(x, t_o)$ so $f_n(x) \rightarrow f(x, t)$. Since $g(x)$ dominates $f(x, t)$ and f_n converges pointwise, by DCT,

$$\lim \int f_n d\mu = \int f(x, t_o) d\mu$$

or equivalently $F(t_n) \rightarrow F(t_o)$. Since $t_n \rightarrow t_o$ is an arbitrary sequence then $\lim_{t \rightarrow t_o} F(t) = F(t_o)$ for all t_o .

ii) Suppose there also exists $t \in [a, b]$ and partial derivative $\frac{\partial f}{\partial t}$ exists such that $|\frac{\partial f}{\partial t} f(x, t)| \leq g(x)$ almost everywhere. Let $d_n(x) = \frac{f(x, t+h_n) - f(x, t_o)}{h_n}$ be a sequence where $h_n \rightarrow 0$. Then by definition of derivative, $d_n \rightarrow \frac{\partial f}{\partial t}$. By assumption, $|d_n| \leq |\frac{\partial f}{\partial t}| \leq g(x)$ a.e. so by DCT,

$$\lim_{n \rightarrow \infty} \int d_n d\mu = \int \frac{\partial f}{\partial t}(x, t) d\mu.$$

Since $\int d_n d\mu = \int \frac{f(x, t+h_n) - f(x, t_o)}{h_n} = \frac{F(t) - F(t_o)}{h_n}$ then $F'(t) \rightarrow \int \frac{\partial f}{\partial t}(x, t) d\mu$. □

2. Prove that $\int_{(0, \infty)} x^n e^{-x} dx = n!$ by differentiating $\int_0^\infty e^{-tx} dt = 1/x$.

Proof. Starting with $\int_0^\infty e^{-tx} dt = \frac{1}{x}$, differentiate both sides such that $\frac{d^n}{dx^n} \frac{1}{x} = \frac{(-1)^n n!}{x^{n+1}}$ and $\frac{d^n}{dx^n} \int_0^\infty e^{-tx} dt = \int_0^\infty \frac{d^n}{dx^n} (e^{-tx}) dt = \int_0^\infty (-1)^n t^n e^{-tx} dt$. Then $\int_0^\infty t^n e^{-tx} dt = \frac{n!}{x^{n+1}}$ so when $x = 1$, $\int_0^\infty t^n e^{-tx} dt = n!$. If $x \in [a, b]$ then $t^n e^{-t} \leq t^n e^{-tx} \leq t^n e^{-ta}$ so the necessary conditions of theorem 1.129 are satisfied and the differentiation under the integral sign is valid. □

3. Prove that if f_n is a sequence of decreasing nonnegative measurable functions with $\int f_1 d\mu < \infty$, then

$$\int \lim_{n \rightarrow \infty} f_n d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu.$$

Proof. Since $\int f_1 d\mu < \infty$ then $f_1 \in L^1(\mu)$ and f_1 dominates f_n as a sequence of decreasing nonnegative measurable functions. Furthermore, $\lim_{n \rightarrow \infty} f_n = 0$ since f_n is bounded below by 0 and decreasing. Thus DCT shows $\int \lim_{n \rightarrow \infty} f_n d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu$. $\square$

4. Let $y_1, \dots, y_M$ be distinct points in $\mathbb{R}$, and $\beta_1, \dots, \beta_M \in \mathbb{R}$. Let

$$f(x) = \prod_{j=1}^M |x - y_j|^{-\beta_j}.$$

Find a necessary and sufficient condition on $(\beta_1, \dots, \beta_M)$ for f to belong to $L^1(\mathbb{R}, m)$.

Proof. $\square$

5. Define $g: \mathbb{R} \rightarrow \mathbb{R}$ by

$$g(x) = \begin{cases} |x|^{-1/3}, & |x| < 1, \\ 0, & |x| > 1. \end{cases}$$

Let $\{r_k\}_{k=0}^\infty$ be an enumeration of all points in $[0, 1] \cap \mathbb{Q}$. Let

$$f(x) = \mathbf{1}_{[0,1]} \sum_{k \in \mathbb{N}} 2^{-k} g(x - r_k).$$

- (i) Show that $f \in L^1([0, 1])$.

Proof. $\square$

- (ii) Show that f is unbounded on every open set (and remains so after modification on a set of measure zero). Show that f is discontinuous at every point in $[0, 1]$ and remains so after modification on a set of measure zero.

Proof. $\square$

- (iii) Let $0 < \varepsilon < 1$. Construct a compact subset K_ε of $[0, 1]$ such that $m(K_\varepsilon) \geq 1 - \varepsilon$ and $f \mathbf{1}_{K_\varepsilon}$ is continuous on K_ε .

Proof. $\square$

6. Let $f \in L^1([0, 2\pi])$.

- (i) Prove that $\lim_{n \rightarrow \infty} \int_0^{2\pi} f(x) e^{-inx} dx = 0$.

Proof. $\square$

- (ii) Prove that for a set $E \subset [0, 2\pi]$ of positive Lebesgue measure $m(E)$,

$$\lim_{n \rightarrow \infty} \int_E \cos^2(nx + \gamma_n) dx = \frac{m(E)}{2},$$

for any sequence γ_n .

Proof.

□

- (iii) Let $F_n(x) = a_n \cos nx + b_n \sin nx$. Assume that the series $\sum_{n=0}^{\infty} F_n(x)$ converges on a set E of positive measure. Show that $\lim_{n \rightarrow \infty} a_n = 0$ and $\lim_{n \rightarrow \infty} b_n = 0$.

Hint: It might help to express $F_n(x)^2$ in the form $(r_n \cos(nx + \gamma_n))^2$, with $r_n > 0$.

Proof.

□